

MANGESH DARKE

San Francisco, CA ◇ hello@mange.sh ◇ mange.sh/linkedin ◇ mange.sh/github ◇ mange.sh

SKILLS

Languages	Python, Go, C, C++, Java, SQL, Swift, JavaScript, HTML/CSS
Networking & Systems	FRR, BGP, DPDK, Linux
Infrastructure	Docker, nerdctl, Kubernetes, Git
Databases & Libraries	Firebase, MongoDB, NumPy, Pandas, Scikit-learn, JUnit, React

EXPERIENCE

IBM

San Jose, CA

Software Engineer

Oct 2024 - Present

Associate Software Engineer

Apr 2022 - Oct 2024

- Develop back-end infrastructure for the Software-Defined Networking (SDN) overlay that powers IBM Cloud's Virtual Private Cloud (VPC), working across both the control plane and data plane of the network stack
- Build and maintain **fabcon**, the network fabric controller, and **iobricks**, the data plane manager — core services that program and manage connectivity across the VPC overlay network
- Implement control-plane routing with FRRouting (FRR) and GoBGP and high-throughput data-plane packet processing with DPDK on Linux-based servers
- Containerize and orchestrate services with Kubernetes, migrating container tooling from Docker to nerdctl
- Independently built an internal team portal (adopted by 15 engineers) using IBM's agentic AI development platform (Bob) — searchable onboarding documentation that cut new-engineer ramp-up time roughly in half, React network-flow diagrams, a development-server reservation system that replaced ad-hoc Slack coordination, and visualizers that reconstruct network topology from tests

GiveTeens20

Fremont, CA

Web Development Intern

Aug 2017 - Aug 2018

- Led a team of 4 interns in redesigning the organization's website to attract new interns and boost website traffic
- Built dynamic UI elements (embedded videos, content carousels) in React.js and modularized the codebase for faster feature delivery
- Increased traffic to the website from potential interns, resulting in 30% gain in number of interns

HIGHLIGHTED PROJECTS

Network Diagnostics MCP Server

- Built a Model Context Protocol (MCP) server in Python exposing network-diagnostic tools (traceroute, BGP route inspection, overlay topology queries) so LLM agents can debug connectivity via natural language
- Added typed tool schemas and an eval harness measuring tool-call accuracy and reliability

CalNow

- Designed a calendar iOS application in Swift using UIKit and Cocoa Touch, backed by a NoSQL Firebase Realtime Database populated by a WebKit web scraper
- Adopted by 25 users to serve as their main student events portal

PintOS

- Implemented system calls, a priority thread scheduler, and a fast file system to bare-bones version of PintOS in C
- Authored technical design documentation to guide architecture and synchronize developer workflows

Spam/Ham

- Built an email spam filter in Python using a Scikit-learn Logistic Regression classifier with engineered text features
- Validated performance through cross validation and hyper-parameter tuning to achieve 94% accuracy

EDUCATION

University of California, Berkeley

2018 - 2020

Bachelor of Arts in Computer Science

Chabot College

2016 - 2018